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Applying XGBoost for Time Series Prediction in Financial Market Data

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Abstract

Time series forecasting is a central theme in financial market, ability to estimate stock prices and trends accurately has a direct impact on investment strategies and risk management decisions. Statistical methods and neural network-based models tend to falter under the nonlinear and erratic nature of financial data. This work is aware of these shortcomings and proposes a new model, Weighted Chameleon Swarm-driven eXtreme Gradient Boosting (WCS-XGBoost), to improve prediction performance in challenging time series cases. Historical stock price data from credible public sources is collected, emphasizing daily closing prices and corresponding technical indicators. The data is normalized, then goes through feature extraction via Principal Component Analysis (PCA) to lower dimensionality while maintaining signal integrity. The predictive engine's central component, WCS-XGBoost, utilizes Chameleon Swarm Optimization to fine-tune XGBoost hyperparameters adaptively, maximizing accuracy and generalization. This framework guarantees that every phase, from raw data aggregation to model training, is optimized for application in financial time series. Metrics like accuracy (98.69%), precision recall, RMSE, and MAPE, outperform traditional models. The suggested system not only offers enhanced predictive power but also presents a scalable solution for market trend analysis and financial decision support. This framework highlights the potential of hybrid evolutionary learning in stock market forecasting methodologies advancement.

Keywords: Financial Time Series, Stock Market Prediction, Financial Data, Weighted Chameleon Swarm-driven eXtreme Gradient Boosting (WCS-XGBoost), Time Series Forecasting.

1 Introduction

Financial markets' variables are complex, non-linear systems influenced by a wide selection of structures, including macroeconomic pointers, geopolitical developments, and institutional behaviour [1]. Time series models are well-suited for capturing these temporal dependencies and evolving dynamics, making them valuable tools for forecasting market trends such as price, returns, volume, and volatility [2-3]. These estimates enable investors, institutions, and regulators to make educated decisions and manage risk more efficiently. Time series prediction is important in financial market analysis, especially for projecting stock values based on previous trends. This process involves analyzing temporal patterns in past data to anticipate future movements

[4]. Two main approaches guide stock market prediction: technical analysis, which focuses on identifying patterns in past prices, and fundamental analysis, which considers economic indicators, company performance, and management quality [5]. However, challenges remain due to the unpredictable and volatile often exhibit co-movement, seasonality, anomalies, and abrupt changes. The foundational assumption of technical analysis, in which patterns tend to repeat over time, underlines the relevance of time series models in modern financial prediction [6-7]. The objective is to enhance an effective time series prediction model for financial market data to enhance stock price forecasting accuracy and support strategic decision-making under uncertainty.

Key contribution

- ❖ The goal is to improve an efficient time series prediction model for financial market data to growth the accuracy of stock price forecasts and facilitate strategic decision-making in the face of uncertainty.
- ❖ Stock market simulation data is preprocessed with z-score normalization for faster convergence and stability. PCA is used for feature extraction and dimensionality reduction.
- ❖ The core framework employs a WCS-XGBoost model used for improving accuracy in nonlinear financial time series.
- ❖ Evaluation shows WCS-XGBoost outperforms traditional methods, demonstrating its robustness for stock market prediction.

The introduction was defined in Section 1, and the related work was detailed in Section 2, the methodological flow was identified in Section 3, the results and discussion were reviewed in Section 4, and the research was concluded in Section 5,

2 Related work

Stock price estimating was improved by integrating media sentiment with financial data in the research [8]. It employed off policy proximal policy optimization to address sentiment class imbalance and used a LSTM model to capture temporal trends. Results enhanced prediction accuracy and better detection of minority sentiments. Limitations included potential underperformance during periods of high market volatility due to insufficient or inconsistently distributed. Research described a novel RedRVFL [9] network for forecasting financial time series. The model's recurrent hidden layers had random initialization and fixed weights, which ensured stability throughout training. It used weighted hidden layers for DL, which extracted complicated patterns from datasets. Comparative result demonstrated greater forecasting accuracy and predictive capabilities. As a result of deep stacked layers and no weight tuning, the model overfit noise in certain volatile financial dataset. The clustering approach utilized auto encoders to improve predictive model performance in the financial industry in the research [10]. The technique, which was applied to prominent financial data demonstrated that using auto encoders improved the granularity and quality of clustering, successfully separating various groups of financial time series. More accurate financial forecasting models are made possible by changes to clustering algorithms, which offer helpful information for risk management and investing strategies. CNN was used to study stock market mood [11], producing accurate risk alerts. While CNN models looked at emotional states over time and their implications for future market sentiment and risk, CNN collected local features from network text input.

The method produced a comprehensive analysis of stock market sentiment and useful early warning indicators of possible issues by addressing order requirements and long-term dependences in time series data. Table 1 shows the summary of recent research on financial marketing.

Table 1: Comprehensive analysis of existing models and techniques in financial prediction

Reference	Objective	Methodology	Findings	Limitations
Li et al., (2024)	increased stock price forecasting accuracy with a new hybrid model.	K-means Clustering group sub-series with similar sample entropy for hierarchical decomposition and prediction.	Outperformed existing models across multiple indices; each component proved effective.	Complexity and computational cost hinder real-time application.
Lazcano et al., (2023)	Improved oil price forecasting accuracy by combining GCN and BiLSTM models.	Developed a hybrid BiLSTM-GCN model for time series prediction.	Achieved lower RMSE, MSE, MAPE, and higher R^2 than individual models and traditional approaches.	Potential computational complexity and need for further validation on diverse datasets.
Albahli et al., (2022)	Stock market moves are forecasted using Google Finance and Twitter data.	Developed an extended opinion lexicon trained with ELM	Achieved 86.06% accuracy, outperforming existing methods.	Limited to Twitter and Google Finance data
Usmani et al., (2024)	Simplified time series forecasting by automating preprocessing and model selection through a framework called <i>Preptimize</i> .	DL techniques to analyze a sample from a dataset and recommend optimized prediction model blueprints.	Preptimize outperformed or matched benchmark models in forecasting tasks across sectors like stock prices, cryptocurrency, and power consumption.	Effectiveness vary with rare or highly domain-specific time series structures not represented in its training or optimization logic.
Mun and Kim, (2025)	examined how to improve investment strategies by using LLM sentiment analysis of financial news.	Sentiment analysis was performed on Nasdaq's top 30 companies using discriminative and generative approach.	Generative LLMs outperformed discriminative models, with Super ICL and Bootstrapping enhancing performance.	Potential risks and data considerations related to LLM explainability and market unpredictability remain areas for

				future exploration.
González-Núñez et al., (2024)	Developed an Artificial Organic Network for stock market prediction using the Index Tracking Problem.	Forecasted eight indices and compared performance with and other ML methods.	Achieved high accuracy (e.g., $R^2 = 0.9806$).	Needs validation across diverse economic scenarios.

2.1 Research gap

The computational and complexity cost constrains real-time usage and, hence, its applicability for high-frequency trading or situations calling for quick decision-making in [12]. The computational complexity of the hybrid BiLSTM-GCN model restrict its usage, and additional verification across varied datasets is required for wider usage [13]. It is based on limited Twitter and Google Finance data that may not effectively encompass the extensive range of factors affecting stock market action and therefore its generalizability [14]. To overcome this issues, WCS-XGBoost offers a promising solution, enhance prediction performance in complex time series scenarios.

3 Methodology

The objective is to create a sophisticated time series prediction model to improve the accuracy of stock price forecasts and facilitate strategic decision-making in the face of uncertainty. Stock market simulation dataset was obtained from the kaggle platform, followed by z-score normalization to standardize feature scales, enhancing training efficiency and stability. PCA is applied for dimensionality reduction, preserving key information while removing noise. WCS-XGBoost framework adaptively fine tunes hyperparameters to capture complex nonlinear financial patterns, improving prediction performance. Overall proposed flow as described in Figure 1.

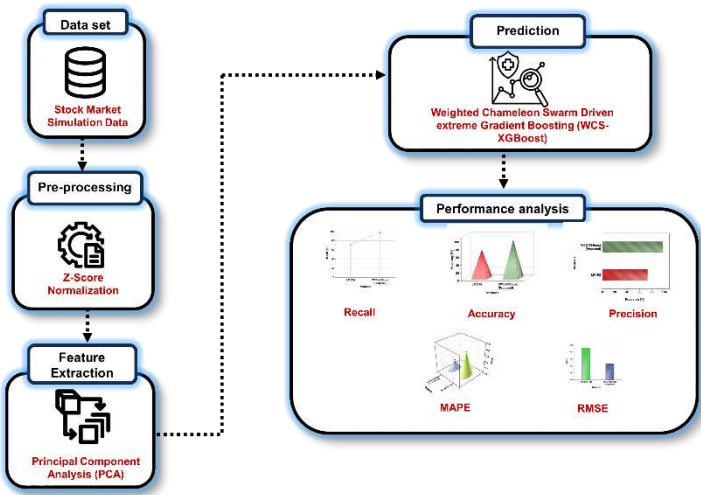


Figure 1: Flow diagram of methodology

3.1 Dataset

The dataset for the stock market simulation is obtained from the Kaggle website [18]. It includes real stock market data generated with Geometric Brownian Motion to model price changes and Markov Chains for forecasting trends. The functionalities comprise forecasting time series data, modelling financials, and simulating algorithmic trading. The data characteristics are shown in Table 2.

Table 2: Data features

Date	Company	Sector	Open	High	Low	Close	Volume	Market_Cap	PE_Ratio	Dividend_Yield	Volatility	Sentiment_Score	Trend
#####	Uber	Technology	100	101.0361	97.47781	100	171958	5.16E+11	24.25322	0.16309	0.047484	0.939232	Bearish
#####	Tesla	Automotive	100.0711	102.038	97.15268	100.0711	196867	9.76E+11	18.60285	0.288515	0.022472	0.469417	Bearish
#####	Panasonic	Finance	99.85783	101.5175	98.1081	99.85783	181932	4.6E+11	10.72893	2.221827	0.019991	0.399193	Bullish
#####	Tencent	Automotive	98.85166	101.3038	96.99822	98.85166	153694	5.58E+11	14.5827	1.37774	0.036166	0.705853	Stable
#####	Wells Fargo	Automotive	98.39112	99.99049	96.23071	98.39112	169879	8.61E+11	37.49111	3.110198	0.03477	-0.76835	Stable
#####	Snapchat	Consumer	99.31567	101.9511	97.61518	99.31567	160268	5.36E+11	28.65938	4.962211	0.023284	0.411912	Bearish
#####	Adobe	Technology	100.5501	102.752	98.64086	100.5501	104886	1.85E+11	22.36167	1.931616	0.047124	0.007377	Bullish
#####	Oracle	Technology	100.9013	103.5124	98.37995	100.9013	187337	3E+11	5.445021	2.869206	0.043794	0.658301	Stable
#####	Novartis	Aerospace	101.0631	104.0734	98.83507	101.0631	137498	3.11E+11	39.73454	2.078019	0.018959	0.016283	Stable

3.2 Data preprocessing using Z-score normalization

After collecting the data, Z-score normalization is applied to continuous characteristics like stock prices, trading volume, and market capitalization. Z-score normalization, which is also referred to as standardization, transforms data to give each characteristic a mean of 0 and a Standard Deviation (SD) of 1 (Equation 1).

$$z = \frac{x - \mu}{\sigma} \quad (1)$$

Where x stands for the initial feature value, μ represents the mean of the feature, σ represents the SD of the feature, and z is the normalized value. Table 3 depicts the data preprocessing outcome and key metrics for model features.

Table 3: Outcome of preprocessed data

Date	Company	Sector	Open	High	Low	Close	Volume	Market_Cap	PE_Ratio	Dividend_Yield	Volatility	Sentiment_Score	Trend
2022-01-01	Uber	Technology	0.0000	0.2220	-0.1774	0.0000	1.3926	0.5537	0.6469	-0.7332	0.1491	1.1154	-1
2022-01-02	Tesla	Automotive	0.0001	0.3225	-0.6155	0.0001	1.7414	1.0419	-0.1787	0.7277	-0.9560	-0.8150	-1
2022-01-03	Panasonic	Finance	-0.0276	-0.0577	-0.0803	-0.0276	0.2516	-0.2975	-0.8866	1.4976	-1.2685	-0.4509	1
2022-01-04	Tencent	Automotive	-0.1327	-0.0895	-0.9753	-0.1327	-0.3941	0.0144	-0.6089	-0.0402	-0.3302	0.7675	0

3.3 Feature extraction using PCA

After pre-processing the data, PCA is employed for feature extraction to reduce dimensionality and highlight the most significant features. It enhances data representation and system analysis by selecting relevant variables from large datasets in stock market data, enabling efficient analysis and extraction for predictive modeling. It reduces the total translation error; this transformation is carried out using PCA by identifying the $\emptyset$ feature of reflecting and vectorizing N -dimensional information into the package. Let's assume that v_s is a random dimensional N with a mean of $(\emptyset)$, representing the input data recordings, Equation (2).

$$\emptyset = \frac{1}{N} \sum_{s=1}^N v_s \quad (2)$$

The definition covariance matrix is denoted in Equation (3), here N is the number of sample data, E is determined by finding eigenvalues and eigenvectors of $(w_s - \mu)(w_s - \mu)^S$. Data factor levels highlight the most influential factors in content consumption.

$$E = \frac{1}{N} \sum_{s=1}^N (w_s - \mu)(w_s - \mu)^S \quad (3)$$

PCA solves the variance matrix DDu_s and an eigenvalue of λ_s concern using Equation (4), a threshold value that ensures the explained variance is sufficiently high. Contextual characteristics represent usage market patterns, including device and skill type.

$$DDu_s = \lambda_s u_s \quad (4)$$

The PCA technique identifies crucial features in eigenvalues and eigenvectors, resulting in a better representation of data.

3.4 Prediction using WCS-XGBoost

WCS-XGBoost approach is applied to enhance time series prediction in financial market data. By applying WCS, the model accounts for dependencies in the data that are both short- and long-term, with XGBoost effectively handling non-linear relationships and interactions. This method improves forecasting accuracy and generalization, making it well suited for predicting complex financial trends such as stock prices or market fluctuations.

3.4.1 EXtreme Gradient Boosting (XGBoost)

XGBoost is used to precisely forecast market trends, like stock prices or volatility, through efficient handling of large datasets and delivering strong, interpretable predictions. Gradient Boosting (GBoost) is less successful in noisy or high-dimensional data since it does not provide probabilistic outputs or uncertainty estimation. XGBoost is used to address difficulties in a GBoost technique with probabilistic regression. It simulates whole probability distributions and provides uncertainty estimates and more precise predictions, all essential for complex, real-time applications in the stock market. In general, XGBoost is a technique that seeks to estimate the parameters of a probability distribution $O_\theta \left(\frac{z}{w} \right)$, $w \in \mathbb{R}$ is the target variable, and $\theta \in \mathbb{R}^K$ is the distribution's parameter vector. The mean and the standard deviation involved assume a normal distribution. Three primary basics make up XGBoost: 1) a parametric probability distribution $e^{(m)}$ 2) the base learners $O_\theta \left(\frac{z}{w} \right)$, and 3) a

suitable scoring rule $K(O_\theta, z)$. Shallow decision trees are used as foundation learners for the XGBoost algorithm in this architecture because of their practical efficacy and efficiency in real-time settings. The logarithmic score is expressed mathematically as Equation (5), which is the most often used scoring rule.

$$K(\theta, z) = -\log(O_\theta(z)) \quad (5)$$

Where $K(\theta, z)$ denotes the cost function, defined as the negative log-likelihood of observing data point, $-\log(O_\theta(z))$ is the negative log-likelihood of z , in statistics and ML. XGBoost uses the natural gradient $\nabla_\theta K(\theta, z)$ to learn the model parameters; it is driven by information geometry. More precisely, under the assumption that have a training set $S = \{(w, z)\}_{j=1}^N$, it begins by estimating a common set of parameters $(\theta^{(0)})$ for the whole training set as follows in Equation (6).

$$\theta^{(0)} = \underset{\theta}{\operatorname{argmin}} \sum_{j=1}^N K(\theta, z_j) \quad (6)$$

Where $\theta^{(0)}$ represents the initial estimate, $\underset{\theta}{\operatorname{argmin}}$ finds the value of θ that minimizes the following expression $\sum_{j=1}^N K(\theta, z_j)$, which denotes a summation over all N data points with relation to the predicted parameters up to that tree, Equation (7).

$$h_j^{(m)} = J_K(\theta_j^{(m-1)})^{-1}, \nabla_\theta K(\theta_j^{(m-1)}, z_j) \quad (7)$$

The optimization formula LF^* represents an optimal solution, and $O_{W,Z}(mg(W, Z) < 0)$ could be an objective function involving the parameters W and Z , which is represented in Equation (8).

$$LF^* = O_{W,Z}(mg(W, Z) < 0) \quad (8)$$

Where $J_K(\theta)$ denotes a generalized Jacobian approximation, depending on the context, $\nabla_\theta K(\theta, z) \nabla_\theta K(\theta, z)^S$ is the outer product of the gradient prediction. The following Equation (9) uses the preprocessed sensor input data to determine the necessary parameters for precise stock market data prediction.

$$o^{(m)} = \underset{o}{\operatorname{argmin}} K(\theta_j^{(m-1)} - o \cdot e^{(m)}(w_j), z_j) \quad (9)$$

Where $o^{(m)}$ represents the optimal step size at iteration m , *again*, this means to find the value of the o , that minimizes the expression following it, $o \cdot e^{(m)}(w_j)$ is a scaled step in the search direction and the settings are modified using Equation (10).

$$\theta_j^{(m)} = \theta_j^{(m-1)} - \eta(o^{(m)} e^{(m)}(w_j)) \quad (10)$$

The XGBoost algorithm enhances financial market time series forecasting by detecting non-linear trends, and offering probabilistic forecasts. It produces more precise, stable, and accurate forecasts, even under high-dimensional data conditions.

3.4.2 Weighted Chameleon Swarm (WCS)

WCS is applied to optimize XGBoost for accurate dynamic predictions in financial markets to utilize its bio inspired swarm intelligence, mimicking a chameleon's adaptive behaviours. Chameleons have super vision, which allows them to locate and follow prey in a variety of situations. It changes colours to match their

environment, traverse through deserts, and climb trees. The system simulates chameleons' natural hunting behaviour, including catching, chasing, and tracking.

Initialization and Function Evaluation: The initialization of WCS and its subsequent implementation are important components of modeling and optimizing predictive models to support stock market development. z_s^j defines its initial population in a search space with s dimensions using $z_{s,1}^j, z_{s,2}^j$ chameleons. Every chameleon shows a solution to the optimization problem of $k_i + q(v_i - k_j)$ has a position at a certain iteration z^j given by Equations (11) and (12).

$$z_s^j = [z_{s,1}^j, z_{s,2}^j, \dots, z_{s,c}^j] \quad (11)$$

$$z^j = k_i + q(v_i - k_j) \quad (12)$$

Chameleon's Eyes Rotation: By rotating their eyes, chameleons determine where prey is located to optimize time series stock market data. The ability to turn aids in 360-degree prey detection; the subsequent actions take place as follows: The initial position of the chameleon is the center of gravity (i.e., the start); the rotation matrix is used to detect the location of the prey; The chameleon's position is updated using the rotation matrix at the center of gravity; the chameleons are brought back to the initial position to optimize market stage. $z_{s+1}^{j,i}$ updates the position at the $s + 1$ iteration and $z_s^{j,i} + O_1(O_s^{j,i} - H_s^i)$ is global solution. $z_s^{j,i} + \mu(v^i - k^i)q_3 + k_a^i$ denotes the positive control parameters, which are presented in Equation (13).

$$z_{s+1}^{j,i} = \begin{cases} z_s^{j,i} + O_1(O_s^{j,i} - H_s^i)q_2 + O_2(H_s^i - z_s^{j,i})q_1 & q_1 \geq O_o \\ z_s^{j,i} + \mu(v^i - k^i)q_3 + k_a^i \text{dgn}(\text{rand} - 0.5)q_1 & q_1 < O_o \end{cases} \quad (13)$$

Hunting of prey: The WCS algorithm's prey behaviour hunting is essential for reducing the distribution of resource deployment for financial market building construction. It optimizes the model parameters, efficiently handling complex market data. More informed and effective decision making in financial strategies can be enabled using Equation (14).

$$u_{s+1}^{j,i} = xu_s^{j,i} + d_1(H_s^i - z_s^{j,i}) + d_2(O_s^{j,i} - z_s^{j,i})q_2 \quad (14)$$

Where, $u_{s+1}^{j,i}$ denotes the updated position of the agent at step $s + 1$ in the swarm that represents the agent's new state after movement. d_1, d_2 are weight factors that control the influence of the prey hunting behaviour. WCS algorithm modifies the location of agents within the swarm, optimizing their waves towards more favorable locations (prey or optimum points), allowing for better resource allocation in financial market strategies. Pseudocode 1 shows the proposed WCS-XGBoost method working procedure.

Pseudocode 1: Weighted Chameleon Swarm-driven eXtreme Gradient Boosting (WCS-XGBoost)

Initialize the parameters $\theta^{(0)} = \text{argmin}_{\theta} \sum_{j=1}^N K(\theta, z_j)$

XGBoost Model function

def XGBoost_model($X_{\text{train}}, y_{\text{train}}, X_{\text{test}}$): $K(\theta, z) = -\log(O_{\theta}(z))$

```

model = xgb.XGBRegressor

model.fit

y_pred = model.predict(X)

return y_pred

# Simulate the WCS parameter (simplified version)

def WCS_update(X_train, y_train):

    # Simulating WCS "hunting" update based on  $z_s^{j,i} + O_1(O_s^{j,i} - H_s^i)q_2 + O_2(H_s^i - z_s^{j,i})q_1$ 

    # Simplified update mechanism  $u_{s+1}^{j,i}$ 

    positions = np.random.randz_s^j = [z_{s,1}^j, z_{s,2}^j, \dots, z_{s,c}^j] # Initialize positions (weights)

    positions -= 0.1 * np.random.randz^j = k_i + q(v_i - k_j) # Simplified update rule

    return positions

# Use WCS to update parameters (simplified)

optimized_params = WCS_update(position)

# XGBoost model with optimized parameters

y_pred = XGBoost_model(X, Y)

# Evaluate the model

End function

```

WCS-XGBoost utilizes a swarm intelligence optimization method to enhance the accuracy of predictions for complex time series scenarios, making the model more suitable, robust, and well-performing in difficult financial forecasting tasks.

4 Result and discussion

The aim is to develop an effective time series prediction model for financial market data to enhance stock price forecasting accuracy and support strategic decision-making under uncertainty. Table 4 shows the experimental setup.

Table 4: Experimental setup

Components	Specification
Operating System	Windows 10
Graphics Card (GPU)	NVIDIA GeForce RTX 3060
Processor (CPU)	Intel Core i7

Software	Python 3.x, pandas, NLTK, TensorFlow/Keras, Flask
Storage	500 GB SSD
RAM	16 GB DDR4

ROC: The ROC curve graphically illustrates how well a classification model performs. To evaluate the discriminating model, the TPR and FPR are compared across various thresholds. Figure 2 shows the outcome of the ROC curve.

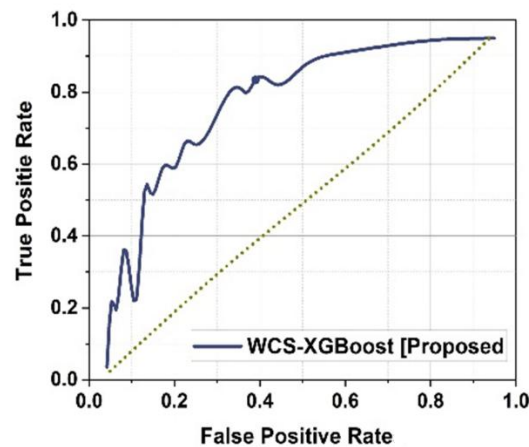


Figure 2: Graphical represent of ROC

As the number of training epochs increases (x-axis), the model's accuracy on the training data (y-axis) improves steadily, indicating that the WCS-XGBoost is effectively learning and enhancing its performance throughout training.

Stock Market Prediction

WCS-XGBoost predicts stocks into predefined categories using learned representations from historical data. It incorporates technical indicators, such as sentiment analysis, and fundamental metrics to enhance performance on financial time series. Figure 3 shows the Stock Market Prediction. This model's close tracking of the actual stock price indicates significant forecasts for stock trends, although slight deviations show the very nature of stock price predictions.

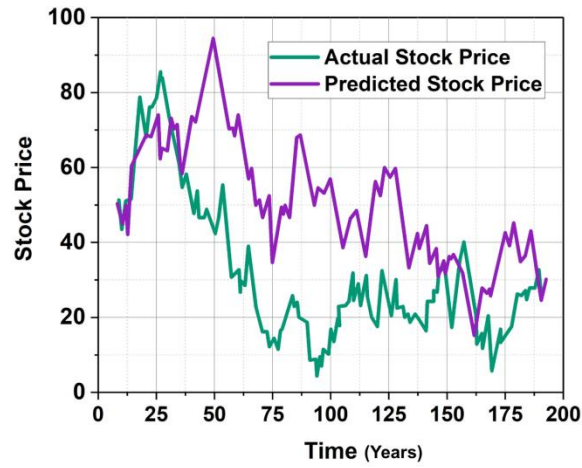


Figure 3: Stock Market Prediction

Comparative analysis

The WCS-XGBoost method is compared with existing methods, including BiLSTM [20], and LR [19], for assessing the effectiveness of the proposed approach. The efficacy of WCS-XGBoost and existing approaches is evaluated using recall, accuracy, precision, RMSE, and MAPE, which is represented in Table 5.

Accuracy is the degree to which a measurement or computation closely matches the real value of observations and corresponds to an accurate value.

Precision is the degree of precision in calculating the potential between measurements of the same element.

Recall is the capacity of a model to identify each significant incident in a specific collection of data. Figure 4 illustrates the result of accuracy.

Table 5: Numerical outcome of metrics

Methods	Accuracy	Precision	Recall
LR[19]	72.7	72.9	72.7
WCS-XGBoost [Proposed]	98.69	98	98

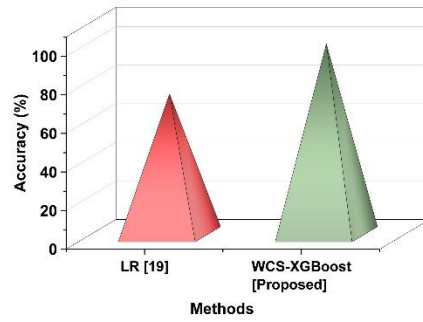


Figure 4: Result of accuracy

WCS-XGBoost achieved an accuracy of 98.69%, while the existing approaches LR attained 72.7% respectively. It demonstrates that the suggested strategy is more efficient than existing methods. Figure 5 shows the outcome of precision.

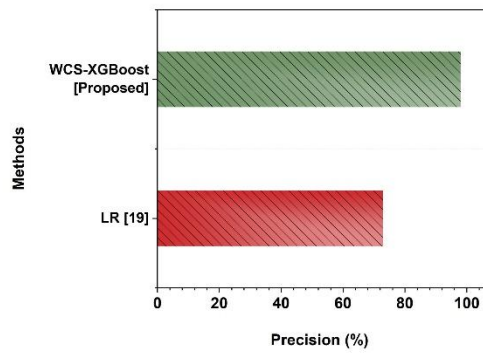


Figure 5: outcome of precision

When compared with existing method, WCS-XGBoost attained a precision of 98%, outperforming LR (72.9%). It demonstrates that the suggested strategy is more efficient than existing methods. Figure 6 depicts the performance outcome of recall.

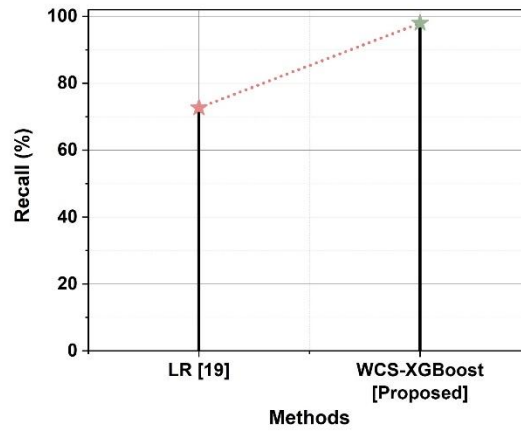


Figure 6: Performance outcome of recall

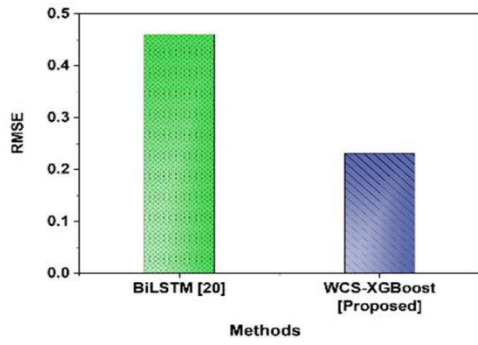
WCS-XGBoost outperformed the existing approach, achieving a recall of 98%, compared to LR (72.7%). It demonstrates that the suggested strategy is more efficient than existing methods.

RMSE calculates the mean magnitude of prediction errors in a model, applying a severe penalty for larger errors due to the squaring of the difference between expected and actual values. **MAPE** refers to the average percentage differences between predicted and actual values, offering relative indicators of forecast accuracy.

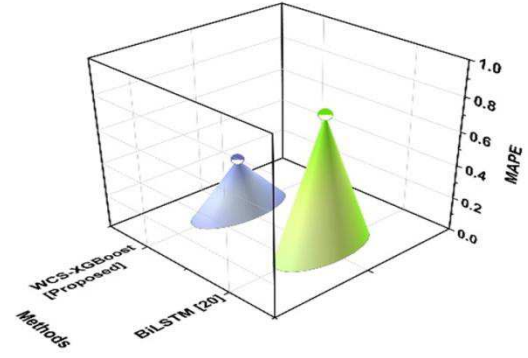
Table 6 shows the numerical result of error metrics. Figure 7 shows the evaluation of (a) RMSE and (b) MAPE.

Table 6: Numerical result of error metrics

Methods	RMSE	MAPE
BiLSTM [20]	0.4605	0.795
WCS-XGBoost [Proposed]	0.2312	0.321



(a)



(b)

Figure 7: Evaluation of (a) RMSE and (b) MAPE

The BiLSTM performed moderately with RMSE of 0.4605, and MAPE of 0.795. The WCS-XGBoost model outperformed both, with much lower RMSE (0.2312) and MAPE (0.321). These findings demonstrate that WCS-XGBoost outperforms existing approaches for scalable solution for market trend analysis and financial decision support.

The research developed an effective time series prediction model for financial market data to enhance stock price forecasting. The WCS-XGBoost system was compared with conventional systems such as LR [19] and BiLSTM [20]. LR [19] has complexity in managing high-dimensional data and was susceptible to kernel regression assortment in stock market. The systems were prone to over fitting and needed considerable training data, limiting their relevance in complex market analysis methods. BiLSTM [20] overfit because of its complexity, particularly when trained on restricted, highly detailed real-world market data. In contrast, the WCS-XGBoost technique efficiently overcomes these restraints by capturing nonlinear connections, directing feature significance, and making more robust and precise stock market forecasts.

5 Conclusion

To enhance the accuracy of stock price prediction and aid in strategic decision-making under uncertainty, the study developed an effective time series prediction model for financial market data. It introduced the WCS-XGBoost model as an effective method for solving the challenges of stock market time series prediction. Once the stock market data is collected, z-score normalization and PCA-based feature reduction are employed. WCS-XGBoost framework heavily improved accuracy and stability of predictions. Experiment results exhibited accuracy (98.69%), precision (98%), recall (98%), MAPE (0.321) and RMSE (0.2312). WCS-XGBoost model provides a generalizable and scalable solution for financial decision support and market trend forecasting. Future research will continue to investigate real-time adaptation as well as integration with multi-source data streams to further enhance forecasting under changing market environments.

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DECLARATION

Ethics approval and consent to participate: I confirm that all the research meets ethical guidelines and adheres to the legal requirements of the study country.

Consent for publication: I confirm that any participants (or their guardians if unable to give informed consent, or next of kin, if deceased) who may be identifiable through the manuscript (such as a case report), have been given an opportunity to review the final manuscript and have provided written consent to publish.

Availability of data and materials: The data used to support the findings of this study are available from the corresponding author upon request.

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Acronyms	Description
TN	True Negative
TP	True Positive

FP	False Positive
FN	False Negative
LSTM	Long-short term memory
RedRVFL	Recurrent Ensemble Deep Random Vector Functional Link
DL	Deep learning
CNN	Convolutional Neural Networks
GCN	Graph Convolutional Networks
BiLSTM	Bidirectional Long Short-Term Memory
ELM	Extreme Learning Machine
LLMs	large language models
TPR	true positive rate
FPR	false positive rate
ROC	Receiver Operating Characteristic